



General Directorate of Health
Division of the cantonal doctor

VAUDOIS PLAN OF PREVENTION AND HEALTH INTERVENTION IN CASE OF HEAT WAVE

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TERMINOLOGY

Heatwave:	There is no universal definition of heatwave, but, general, we admit that it designates a heat wave which, by its duration, its intensity or aggravating factors (night temperatures high, high ozone) is likely to cause serious health consequences, including death, particularly among vulnerable people.
Vulnerable people:	individuals who, for reasons related to age, the presence of disease(s) chronic or physical activity, suffer more from the effects of heat on health than the rest of the population.
Harvesting effect:	The “Harvesting effect”, or “mortality displacement” corresponds to an increase in the mortality rate linked to environmental phenomena (pollution, heatwave, etc.) followed by a drop in the mortality rate. The hypothesis is that events exceptional events (pollution peaks, heatwaves, etc.) primarily affect the most vulnerable individuals who would have died within weeks or the following months. For example, of all the victims of the summer 2003 heatwave in France, more than 2/3 had an expectation of residual life of one year and were destined to disappear in 2004, year characterized by a significant deficit in deaths, undermortality increases extending until 2007.
Urban heat island effect:	This term refers to the rise in atmospheric temperatures and surface area in urban centers compared to rural regions neighboring areas. This local human-generated condition is due to a greater heat absorption by urban materials (e.g. concrete, brick, asphalt), reduced convective cooling natural environment caused by the presence of tall buildings, and lack of green space which reduces evaporative cooling. The air temperature in the evening can therefore be up to 12°C higher. Climate change is likely to intensify the island effect urban heat in metropolitan areas. The effect could however be reduced by increasing the reflectivity of urban surfaces, by increasing green spaces in a community through the development of urban forests, parks or green roofs and vertical gardens, spacing new constructions so as to create corridors.

LEGAL BASIS

- ⁹ Civil Protection Act (LProp), 2004
- ⁹ Public Health Act (LSP), 1985

ABBREVIATIONS

AVASAD:	Vaud association for home help and care
AVOP:	Vaudois association of private organizations for people in difficulty
THIS:	Board of state
CHUV:	Vaud university hospital center
CMS:	Medical-social centers
CORE:	Coordination of Vaud healthcare networks
CTMG:	Telephone center for doctors on call
DGE:	General Directorate of the Environment
DGS:	General Directorate of Health
DSAS:	Department of Health and Social Care
EMCC:	Cantonal driving staff
EMS:	Medical-social establishments
FEDEREMS: Employers' federation of Vaudois EMS	
FHV:	Federation of Vaud hospitals
FUS:	Emergency Health Foundation
HEL:	Lausanne Children's Hospital
HéviVA:	Vaud association of medical-psycho-social institutions
HI:	Heat index
MC:	Cantonal doctor
MeteoSwiss: Federal Office of Meteorology and Climatology	
OAJE:	Children's day care office
ORCA:	Disaster organization
ORPC:	Regional Civil Protection Organization
PCI:	Civil protection
RS:	Care networks
SCL:	Municipalities and housing service
SSCM:	Civil and military security service
SVM:	Vaud Society of Medicine
Unity:	University Center for General Medicine and Public Health

PART I

1. INTRODUCTION: CONTENT AND SCOPE OF THE REPORT

The Vaud health prevention and intervention plan in the event of a heatwave (hereinafter heatwave plan) is intended for health professionals and stakeholders involved at one time or another during the surveillance period which extends between June 1 and August 31.

The main objective of the heatwave plan is to prevent and mitigate damage to the health of the population due to extreme heat and avoid overloading the health system.

To do this, the heatwave plan is structured around three axes:

- Information to the general population, professionals in the socio-health sectors as well as those in contact with the most vulnerable people, on the harmful effects of heat on health and on the precautionary measures to be taken during the summer and, above all, during periods of high heat.
- Monitoring of meteorological and socio-sanitary indicators during the period from 1 June to August 31, which allows you to anticipate a heatwave period and act quickly in the event of a problem.
- Coordination of the different actors concerned by a possible health response in the event of severe heatwave.

Since the development of the "Report on the severe heatwave. Health operational plan for the canton of Vaud, Working document – Situation as of July 14, 2009", definitions and implementation procedures work has evolved.

The Cantonal Physician's Division considered it necessary to update the 2009 Report in adapting it to practice and simplifying it so that it is easier and more fluid to read. It was also considered useful to deepen the epidemiological information related to the heatwave which was lacking in the previous version.

The plan proposes a theoretical part which illustrates the essential points linked to the heatwave: definition (to our latitudes), health consequences, risk groups, prevention measures, etc. These are aspects which make it possible to understand the need, for the State, to have a plan in order to protect its population.

The second part of the Heatwave Plan focuses on operational aspects, in particular the actors with respective roles and responsibilities, indicators to monitor and tools to do so, alert levels and health responses.

2. REASONS TO PREPARE FOR THE POSSIBILITY OF A HEAT WAVE

The year 2003 is destined to remain in public health memories and treatises because of a particularly hot and, above all, very deadly summer.

The heat waves recorded during the summer of 2003 caused a significant increase in the number of deaths not only in France (15,000 deaths), but also in several other European countries, especially in urban centers. In Switzerland, mortality during the heatwave increased by 7.5%, or 1,000 deaths, compared to previous summers.

The exceptional nature of the heatwaves of 2003 is destined to fade in favor of regularity and of a greater frequency of this type of event, and this because of ongoing climate change.

On the other hand, the relationship between high temperature and increased mortality and morbidity is established and uncontested. Later in this report this relationship will be discussed in depth. Human beings adapt both physically and through changes in behavior and technical measures (construction methods, urban planning, etc.) to temperatures ambient temperatures commonly higher. This adaptation to new conditions, however, requires several years or even decades¹.

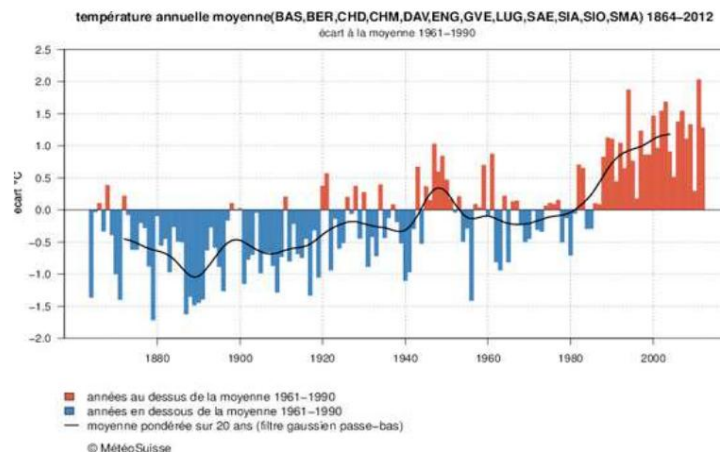
This is why a heatwave plan makes perfect sense. It makes it possible to anticipate the harmful consequences of the heatwave on health, via adequate information to the population and preventative measures; and promotes good responsiveness of all stakeholders concerned in the event of of worsening of the situation on the basis of an early warning system.

¹ Climate change and Switzerland in 2050, OcCC, ProClim, 2007
PLAN , DGS, 03.11.2014-Updated 06.22.2020 VD HEAT WIND

3. CHARACTERIZATION AND TRENDS OF HEAT WAVES

3.1 CLIMATE CHANGE²

Witness to ongoing climate change, the graph below shows the evolution of temperatures in Switzerland since the beginning of systematic measurements in 1864. As a result, from the 1980s onwards, we observed an increase in average temperatures to seasonal levels significantly higher. These differences concern all seasons.



According to the Swiss climate scenarios report CH2011³, we expect a change significant of the Swiss climate during the 21st century. Average temperatures will most likely increase by several degrees in all seasons.

We must expect periods of summer heat and heatwaves that are more frequent, more intense and longer, while the number of cold winter days and nights is expected to decrease.

3.2 DEFINITION OF HEAT WAVE

A heatwave is a heat wave characterized by abnormally high temperatures. high, linked to significant warming of a vast territory, lasting several days or weeks. In Switzerland a heatwave warning is issued when it is forecast, for 3 days consecutive at least, a heat index (Heat index=HI) greater than 90.

This criterion, defined by the American meteorological service and universally used, is calculated from of a complex formula which involves the temperature expressed, in degrees Fahrenheit, and the humidity relative, in percentage. Heat also depends on wind speed and radiative load.

In our latitudes, we observe that relative humidity is very variable below 30°C, but that this parameter is relatively constant during extreme heat (the relative humidity then varies from 20% to 25%). In a more simplified manner, we can say that during periods of extreme heat, the highest HI in a day corresponds to the time of day when the observed temperature reaches its maximum, the effect of relative humidity being negligible.

² www.meteosuisse.ch

³ Climate change scenarios in Switzerland CH2011, September 2011

If the criteria for assessing a situation are based on HI, this concept is not easy for the population to understand; for this reason, when communicating with the outside world, it is preferred to use maximum temperatures of 33-34°C, corresponding, in our latitudes, to HI 90.

MeteoSwiss issues a degree 3 danger warning when the HI is above 90 for at least 3 consecutive days. Since summer 2012, it has introduced a degree 4 danger notice when the HI is above 93 (corresponding to a maximum temperature exceeding 35-36°C) for at least 5 consecutive days. This new criterion has statistical and informative interest allowing MétéoSuisse to distinguish and characterize the types of heat waves as well as their periodicity.

Based on these criteria, it is possible to classify the summers of recent years as follows:

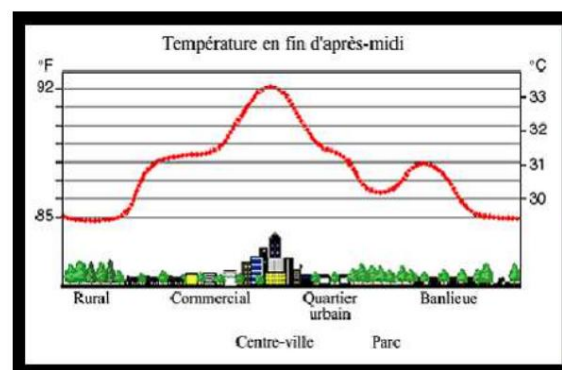
⁴:

Canicule	Jours Tmax > 30°C (*) consécutif	Danger Inférieur Degré 3 (cas)	Danger Degré 3	Danger Degré 4
Août 2012	5	(X)	X	
Août 2011	7	(X)		
Juillet 2010	9	(X)		
Juin 2010	2	X		
Août 2009	8	(X)	X	
Juillet 2006	12		X	
Août 2003	13		X	X
Juillet 2003	9 (7*)		X	
19-30 Juin 2003	10 (8*)		X	(pas exclu)
7-18 Juin 2003	8 (5*)			X

In our regions, heatwaves are accompanied by relatively dry air. This heat situation is better tolerated than when air humidity is high.

Through the "urban heat island effect" (see page 3), daytime and nighttime temperatures result higher in the cities than in the countryside. This local human-generated condition is due to greater heat absorption by urban materials (e.g. concrete, brick, asphalt), the reduction in cooling by natural convection caused by the presence of tall buildings, and the lack of green spaces which reduces cooling by evaporation.

It is for this reason that heat waves cause more deaths in urban centers than in rural areas, peripheries and countryside. The image below illustrates this phenomenon.



Source: Lawrence Berkeley National Laboratory, 2000.

⁴ Lionel Fontannaz, MeteoSwiss, DGS,

4. HEAT WAVE AND HEALTH

4.1 THE EFFECTS OF THE HEAT WAVE ON HEALTH

Weather conditions that may cause health risks depend on the sensitivity, acclimatization and the capacity of the population to assume appropriate behaviors to the intense heat situation – that is, vulnerability to heat.

Certain factors make people more vulnerable than others and are therefore more exposed to health risks.

Intense heat can have several health consequences, from the mildest to the most serious, such as death, which can be avoided. But these aspects are a reality, especially when people were overexposed by neglecting to adapt their behavior to the heat or by not benefiting from adequate support.

Above are the main health effects in descending order of severity.

⁵:

Heat stroke	Of all types of heat-related illnesses, heat stroke is the most serious. It is caused by too high a body temperature. It is defined by an increase in central body temperature greater than 40°C associated with damage to the central nervous system occurring in a context of excess heat in the environment. There are two types of heat stroke: • Classic – little or no sweating; usually occurs in children, people with chronic illnesses, and older adults. • Exertion-induced – is accompanied by an increase in body temperature due to strenuous exertion or workplace exposure in combination with heat; there is generally sweating.
Heat exhaustion	Caused by excessive loss of water and salt. May be accompanied by the following symptoms: heavy sweating, weakness, dizziness, nausea, headache, diarrhea and muscle cramps.
Heat syncope (heat-induced fainting)	Caused by loss of body fluids through sweating and a drop in blood pressure. Symptoms include temporary dizziness and fainting caused by insufficient blood flow to the brain.
Heat cramps	Caused by a salt imbalance which results from the body's inability to replace salts and minerals lost through sweating. The symptoms are sharp pains in the muscles.
Heat rash (miliaria rubra)	The result of inflammation of blocked sweat glands, which is accompanied by small red spots on the skin; may give a tingling sensation.
Heat edema	Swelling caused by heat, usually in the ankles, feet and hands, most often in people who are not regularly exposed to heat.

4.2 PEOPLE AT RISK

From a health point of view, the major problem during a heatwave is represented by the number of deaths premature babies of people suffering from chronic illness(es). Indeed, epidemiological studies show that the majority of heat-associated deaths do not show the classic signs of thermal pathology.

Within the population, there are groups that are more vulnerable than others to heatwaves. If prevention messages must be disseminated and be able to be understood by the entire population in order to

⁵ Adapting to periods of extreme heat: guidelines for assessing health vulnerability, Health Canada, 2011
PLAN , DGS, 03.11.2014-Updated 06.22.2020 VD HEAT WIND

To avoid the pathologies mentioned above, particular attention must be paid to particularly vulnerable people:

People with reduced mobility living at home	Confined to bed/armchair, sometimes dependent on those around them for care, unable to go out at least once a day to cool off or participate in social exchanges.
The elderly	The largest vulnerable group identifiable in advance (decreased heat tolerance, late feeling of thirst, reduced sweating, impaired thermoregulation, co-morbidities, dementia, poly-medications, functional limitations, 'dependence' including drinking, social isolation, possibly housing poorly insulated from heat, etc.).
Patients with chronic diseases including psychiatric	In particular: psychiatric illnesses (including depression) affecting perception of the environment and oneself, drug dependence, use of tranquilizers, alcohol consumption; ailments chronic diseases such as diabetes, cardio-respiratory, cerebrovascular, renal and metabolic diseases. Specific medications can also represent a risk linked to heat (pharmacological effect per se or by dilution/elimination, changes in pharmacokinetics or alteration by storage in heat).
Socially disadvantaged people of all ages:	For example living alone, on low income or in poverty, on the street, etc.
Children < 10 years, but in particular < 5 years	Different metabolism, still immature thermoregulation, dependence on adults including. for drinking, surface-volume ratio higher than in adults, particularly in those < 1 year old.
Outdoor workers	Construction sites, others depending on professional field (e.g. working in compulsory tracksuits, sometimes waterproof, including emergency personnel).

4.3 AGGRAVATING FACTORS

The previous table mentions, in the different categories of vulnerable people, the factors of a biological, physical or socio-economic nature that may constitute an additional risk for health in the event of a heatwave.

There are also other aggravating factors of an environmental nature, such as temperature night time and the ozone concentration rate.

The nighttime minimum temperature is a byproduct of the maximum temperature. When this last reaches 33°C, the night temperature is, in almost all cases, higher than 17°C and, in urban areas, it is between 19° and 21°C. This type of situation therefore does not allow no cooling of the air during the night.

Ozone⁶ is a natural constituent of the atmosphere. Between 10 and 50 km altitude, the ozone layer protects us from the sun's UV rays. But on the earth's surface, ozone results from activities human. Motorized transport, industry and the use of solvents lead to releases of nitrogen oxides and volatile organic compounds, which are transformed into ozone under the effect of strong sunlight. This is why ozone problems appear in spring and summer. One too many high concentration of this gas has harmful effects, both for living beings and for vegetation

⁶ <http://www.vd.ch/themes/environnement/air/polluants-atmospheriques/ozone/>
PLAN , DGS, 03.11.2014-Updated 06.22.2020 VD HEAT WIND

Among the various health consequences are: irritation of the mucous membranes, eye irritations, amplification of reactions to other irritants or allergens (pollens, mites, etc.), difficulties respiratory.

4.4 RELATIONSHIP BETWEEN HEAT WAVE AND MORTALITY

The heatwave of 2003 was striking in its intensity and duration. However, well before this episode, other heatwaves had made headlines due to their health impact and a significant increase in mortality compared to seasonal norms: +96% in Athens in July 1987; +107% in Lisbon and +89% in Seville in June 1987; +150% in Chicago in July 1995⁷.

According to scientific literature, heatwaves which occur at the beginning of summer, unlike Episodes that occur later in the season generally result in higher rates of morbidity and mortality, partly due to a lack of acclimatization. This phenomenon occurs through gradual exposure to heat leading to physiological changes that allow a person to better adapt to heat. The body's capacity for acclimatization may, however, be limited in certain vulnerable populations⁸.

The relationship between temperature and mortality has a U, J or V shape with an optimum thermal, for which mortality is minimal, which is between 15 and 25°C depending on the European country. Mortality increases (from 0.7 to 3.6%) when we move away from this optimal temperature, whether towards high or low temperatures⁹.

When a region-specific temperature threshold is exceeded, mortality increases markedly.

Figure 1 below shows the relationship between mortality and the temperature experienced on the northern slope and the southern slope of the Alps from 1990 to 2003. The regional difference appears clearly: the temperature "optimal", presenting the lowest mortality, is 3.5°C higher on the southern slope of the Alps than on the north slope¹⁰.

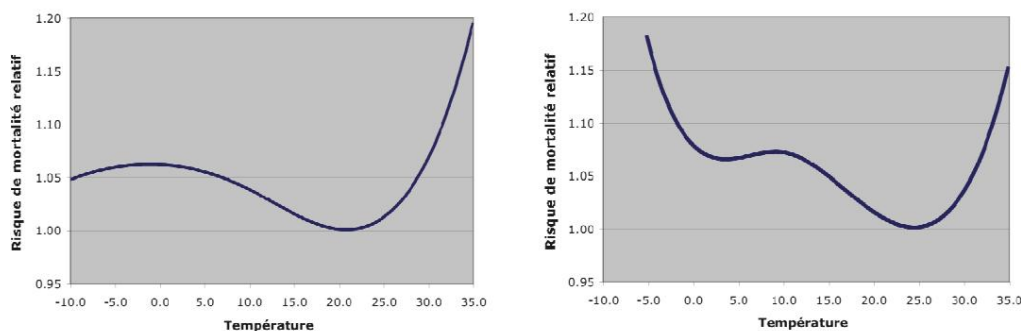


Figure 1: Relation entre mortalité et température sur le versant nord (à gauche) et le versant sud (à droite) des Alpes de 1990 à 2003. Le domaine de température présentant la plus faible mortalité se situe aux environs de 21 °C sur le versant nord des Alpes, soit en gros 3.5 °C au-dessous de celui sur le versant sud des Alpes, qui se situe autour de 24.5 °C. (Source: ISPM Bâle)

Generally, the mortality curve begins to rise at the start of the heat wave, reaching its peak within 72 hours (3 days) after the first temperature peak. The curve subsequently bends more or less quickly, independently of the temperatures.

⁷ Heat waves and mortality in large urban areas. J.-P. Besancenot, Environment, Risks & Health. Sept.-Oct. 2002

⁸ Adapting to periods of extreme heat: guidelines for assessing health vulnerability, Health Canada, 2011

⁹ Cold waves and health in mainland France. Impact, prevention, opportunity of an alert system. InVS, 2009

¹⁰ Climate change and Switzerland in 2050, OCCC, ProClim, 2007.

Excess mortality due to high temperatures is difficult to interpret because it must take into account evolution over several years. Only this type of analysis makes it possible to identify whether it is a long-term effect, term or harvest effect:

- We see a long-term effect when the heat wave induces excess mortality which is not compensated by a drop in the number of deaths in the following period and determines a significant increase in mortality not only during the summer, but also at the balance sheet level annual.
- While the harvesting effect (or Harvesting effect) corresponds to an increase in the mortality rate linked to environmental phenomena (pollution, heatwave, etc.) followed by a reduction in the mortality rate. The hypothesis is that exceptional events (pollution peaks, heatwaves, etc.) primarily affect the most vulnerable individuals who would have died in the following weeks or months. For example, of all the victims of the summer 2003 heatwave in France, more than 2/3 had a residual life expectancy of one year and were destined to disappear in 2004, a year characterized by a significant deficit in deaths, undermortality extending until 2007.

4.4.1 MAIN CAUSES OF MORTALITY

The scientific literature on the events of the summer of 2003 has revealed that despite the existence of certain geographical differences, the main causes of death linked to the heatwave were quite homogeneous within different European cities and can be attributed to respiratory, cardiovascular and cerebrovascular problems¹¹.

Table 2: City specific estimates of the effect of heat waves on daily mortality (% increase and 90% CI) by cause of death among people aged 65+ years.

City	All natural deaths		Respiratory deaths		Cardiovascular deaths		Cerebrovascular deaths	
	% increase	(90% CI)	% increase	(90% CI)	% increase	(90% CI)	% increase	(90% CI)
Athens	21.6	(18.5 - 24.8)	34.5	(24.6 - 45.2)	28.4	(24.0 - 33.0)	33.0	(25.9 - 40.4)
Barcelona	15.6	(11.0 - 20.4)	41.3	(26.4 - 57.9)	21.4	(14.0 - 29.4)	25.1	(12.4 - 39.3)
Budapest	21.1	(17.3 - 24.9)	20.6	(2.6 - 41.7)	24.1	(19.3 - 29.1)	24.6	(15.6 - 34.4)
London	10.4	(8.6 - 12.2)	18.0	(13.4 - 22.8)	9.3	(6.6 - 12.1)	10.6	(5.2 - 16.3)
Milan	33.6	(28.5 - 39.0)	92.5	(72.3 - 115.1)	39.2	(31.2 - 47.6)	49.8	(35.6 - 65.6)
Munich	7.6	(3.8 - 11.5)	3.9	(-0.8 - 30.8)	8.2	(2.8 - 13.9)	14.7	(2.4 - 28.6)
Paris	11.4	(10.0 - 12.9)	27.7	(19.4 - 36.6)	12.3	(8.5 - 16.2)	19.7	(12.1 - 27.8)
Rome	26.8	(23.4 - 30.4)	66.9	(51.9 - 83.3)	37.8	(32.5 - 43.3)	48.0	(38.0 - 58.8)
Valencia	8.5	(1.2 - 16.3)	32.4	(9.1 - 60.7)	20.1	(7.9 - 33.6)	1.4	(-17.4 - 24.4)

4.4.2 PLACE OF DEATH

The table on the next page shows the ratios between observed and expected mortality by location. death (home, hospital, private clinic, EMS, public highway) for the whole of France metropolitan area from August 1 to 20, 2003. The observed mortality was significantly higher than the mortality expected mainly in EMS (Retirement Home), followed by Home and hospitals (Hospital).

¹¹ "The impact of heat waves on mortality in 9 European cities: result from the EuroHEAT project", Environmental Health 2010, 9:37
PLAN , DGS, 03.11.2014-Updated 06.22.2020 VD HEAT WIND

<i>Place</i>	<75 years old	>75 years old	Total
	Ratio of mortality (O/E)	Ratio of mortality (O/E)	Ratio of mortality (O/E)
<i>Home</i>	1.4	1.9	1.7
<i>Hospital</i>	1.2	1.6	1.5
<i>Private Clinic</i>	1.0	1.3	1.2
<i>Retirement Home</i>	1.8	2.0	2.0
<i>Outside</i>	1.0	1.0	1.1

Source: INSERM (O = number of deaths Observed ; E = number of deaths Expected)

4.5 PREVENTION MEASURES

Given the consequences that the heatwave can have on health, which can even lead to death, and taking into account that the worsening of an underlying illness may be more common than direct effects of heat, it is important to remind the population of the main prevention measures.

4.5.1 INDIVIDUAL MEASUREMENTS

These measures are aimed at the entire population in order to avoid the negative consequences of the heatwave. They are simple measures that anyone can apply:

- ÿ Protect the accommodation against the heat (close the shutters, lower the tent fabrics, open parasols) and cool it at night (open blinds and shutters, create drafts, etc.).
- ÿ Do not expose yourself to heat, especially during the hottest hours.
- ÿ Do not leave children in a vehicle parked under the sun.
- ÿ Drink regularly (min. 1.5 liters of water unless there are medical contraindications), avoid alcohol, and eat cold, light foods rich in water (fruits, vegetables, ice cream).
- ÿ Cool down several times a day (shower, damp cloths, air-conditioned or cooled rooms, walk in the forest, swimming pool, etc.). Electric fans can provide temporary relief, but they do not prevent health consequences due to heat.
- ÿ Wear light (cotton), loose clothing and, preferably, light colors.
- ÿ Avoid physical activity, especially during hot hours.
- ÿ Help others (vulnerable populations).
- ÿ Know the information and emergency numbers.
- ÿ Seek the advice of your treating physician, in particular for the adaptation of treatment in the event of chronic disease sensitive to heat waves.

4.5.2 COLLECTIVE MEASURES

Collective measures are issued by the authorities and are aimed either at individuals, citizens, either to public or private communities.

- ÿ Carry out an early and repeated awareness and education campaign for the general public public about risks and protection strategies.
- ÿ Solidarity: check in regularly with isolated friends, relatives and acquaintances, report them to appropriate services where applicable.
- ÿ Actively search for vulnerable residents.
- ÿ Distribute water to homeless people.
- ÿ In care and accommodation institutions (hospitals, EMS, outpatient clinics, socio-economic establishments educational offices, medical offices, etc.): arrange the premises (e.g. adding blinds) or procedures (drink rounds, adaptation of infusions, air conditioning of intensive care and emergencies, etc.).
- ÿ In the long term: build or develop housing and public places compatible with a heatwave (reflective paint on roofs and walls, interior and exterior vegetation, interior fountains and exteriors, hangings on exposed facades, etc.), reduction of greenhouse gas emissions tight.

4.6 THERAPEUTIC MEASURES

Therapeutic measures depend on the severity of heat symptoms and are detailed in the document "Heatwave. Information intended for health professionals" available at the address www.vd.ch/canicule.

In any case, in the presence of warning signals (weakness, confusion, dizziness, nausea, cramps muscular) you must call the CTMG or directly 144 if necessary.

In the event of heat stroke and before help arrives:

- ÿ Refresh the person (shade, damp sheet, possibly ice packs except in children, under ventilation, with massage to compensate for shivers), give them something to drink (never alcohol) if they are well aware.
- ÿ No febrifuges (Paracetamol, Aspirin), which are ineffective for this pathology.
- ÿ If the person loses consciousness, place them in a side safety position.

PART II

5. THE HEAT WAVE PLAN

5.1 OBJECTIVE AND BASIS OF THE HEAT WAVE PLAN

The main objective of the Heatwave Plan is to prevent and mitigate damage to the health of the population. due to the oppressive heat and to avoid overloading the health system.

To do this, the Heatwave Plan is structured around three axes:

- Prevention, which consists of informing the general population, professionals in socio-health environments as well as those in contact with the most vulnerable people, about the harms heat on health and on the precautionary measures to take during the summer and, above all, when periods of extreme heat.
- Monitoring of meteorological and socio-sanitary indicators during the period from 1 June to August 31, which makes it possible to anticipate a heatwave period and act quickly in the event of issue.
- Coordination of the different actors concerned by a possible health response in the event of severe heatwave.

5.2 THE ORGANIZATION

The Heatwave Plan is a health system for which the DGS is responsible for organizing procedures, coordination of actors and dissemination of information. The DGS relies on the expertise of Unisanté for the modeling of the indicators used for heatwave monitoring and on MétéoSuisse for the interpretation of meteorological data.

The system also provides for the active participation of different socio-health actors and partners. Vaudois responsible for the dissemination of information within their own corporation regarding the health situation and the implementation of the various measures provided for in the system, according to the document.

All these actors, as well as other services of the Vaud cantonal administration concerned by the heatwave risk, sit within the heatwave alert group which meets twice a year (before and after summer) and if necessary during the monitoring period.

5.3 THE RESPONSIBILITIES OF THE DIFFERENT ACTORS

In this organization, each actor has their own responsibilities which are summarized below:

- ÿ DGS: the office of the cantonal doctor is responsible for ensuring the conduct of the heatwave plan, it is also responsible for the organization and coordination of measures and actors in link with the Heatwave Plan.
- ÿ MétéoSuisse: is responsible for transmitting meteorological information to the DGS and for provide the necessary support for the interpretation of the forecast.
- ÿ Unifying: its role is to provide technical competence and statistical expertise to the DGS

for the modeling of health and meteorological indicators, updating and management of the database relating to these indicators.

- ÿ **EMCC:** it ensures, for the benefit of decision-making bodies, technical meteorological monitoring and establishes the synthesis of technical monitoring of the different areas concerned by the heat wave. The EMCC also ensures operational availability in the event of the establishment of an Infra ORCA (orange alert) or ORCA (red alert) system (see section 4.2).
chapter 5.7).
- ÿ **DGE:** it monitors ozone concentrations and has appropriate measures
hoc to limit excessive values. Ozone may be a factor
worsening the state of health of the population during heatwaves, this information is integrated into the risk analysis.
- ÿ **Institutions** CHUV, FUS, FHV, FEDEREMS, HévivA, CORES, AVASAD, AVOP, SVM: are called to disseminate, within their organization, any useful information from the DGS relating to the health situation. Their role is also to provide the DGS with any information that could help it assess the situation. They are responsible for the
conduct and coordination of activities within the institutions they represent, they
are therefore asked by the DGS to trigger their ad hoc heatwave plan.

During the heatwave monitoring period, the CHUV, the HEL and the FUS transmit daily information relating to health indicators to the DGS (see section 4.1).
chapter 5.6).
- ÿ **SCL:** The Prefect is responsible for ensuring the relationship between the State and the municipalities for all questions relating to the heatwave. He or she is responsible for transmitting to the municipalities via the Prefects the decision of the head of DSAS in the event of activation of the plans heat wave.
- ÿ **OAJE:** This office is responsible for transmitting to the network of day care structures of the early childhood information relating to preventive measures to be adopted in the event of a heatwave, to respond to any requests and, in the event of extreme heat,
to disseminate information from the DGS on the situation and precautions
take.

5.4 THE MAIN PHASES

The Heatwave Plan is a system which provides for different phases structured around the health monitoring period between June 1 and August 31. The table below illustrates the tasks of partners:

PUBLIC HEALTH SERVICE MEMBERS OF THE ALERT GROUP HEAT WAVE AND OTHERS		
BEFORE THE EVE	- Define the schedule of actions to undertake before the start of the monitoring period.	- Update, if necessary, the respective intervention plans in the event of a heatwave, inform employees of their own institution about the prevention measures to be taken and plan the actions to be undertaken (e.g. for municipalities: recruitment and training of
	- Prepare monitoring tools, in particular by asking Unisanté to update the database with meteorological and socio-health indicators	

	to be continued.	volunteers for visits; identification of elderly people at risk).
	- Develop the communication plan and materials for the various recipients (general population, socio-health professionals, media, etc.). - Update the plan if necessary heatwave based on new scientific foundations. - Make the necessary modifications to the heatwave device based on the evaluations	- Plan their presence during summer in order to guarantee the transmission of information to and from the DGS.
DURING THE EVE (June 1- August 31)	- Capture and monitoring of meteorological and socio-health indicators (see next chapter) - Regular consultation with MétéoSuisse (forecasts, weather trends), Unisanté (data interpretation), DGE (ozone concentration) and canton of Geneva (decision sharing). - In the event of a heatwave, information regularly to members of the Heatwave Alert Group to inform them about the situation and its evolution. - In the event of a persistent heatwave (orange alert level): convening of the heatwave alert group and, if necessary, activation of the heatwave plans after decision by the Head of DSAS. - Development and distribution of press release(s). - Transmission, exchange information with EMCC.	- CHUV, HEL, CTMG, Civil Status Offices, MétéoSuisse: daily transmission of data to the DGS. - Regularly take note of the situation within their own institution or their members (if umbrella) in order to be able to inform the DGS if necessary. - Dissemination of information received by the DGS. - Based on communication from the DGS, triggering of institutional and municipal heatwave plans and activation of the hotline managed by the CTMG. - If the red alert level is reached: overall management of the crisis by the EMCC (health and non-health coordination); management of the health crisis remains the responsibility of the DGS.
AFTER THE EVE	- Assessment of the monitoring period and organization of the Heatwave Alert Group session for the restitution of information.	- Report to the DGS and other partners of their own experience.

5.5 THE MONITORING SYSTEM¹²

Generally speaking, taking into consideration a health risk is based on an organization of health monitoring including:

• surveillance systems which are based on the regular collection of structured information with production of indicators defined a priori;

• a monitoring device allowing the collection and analysis of different signals.

The main purpose of a surveillance system is to deploy a rapid response in the form of actions proven to be effective, in order to limit the impact of the threat on the health of the population, particularly the most vulnerable people.

In this sense, a good articulation between the concepts of “surveillance”, “alert” and “health response” is essential and its operation must be understood and accepted by the parties concerned.

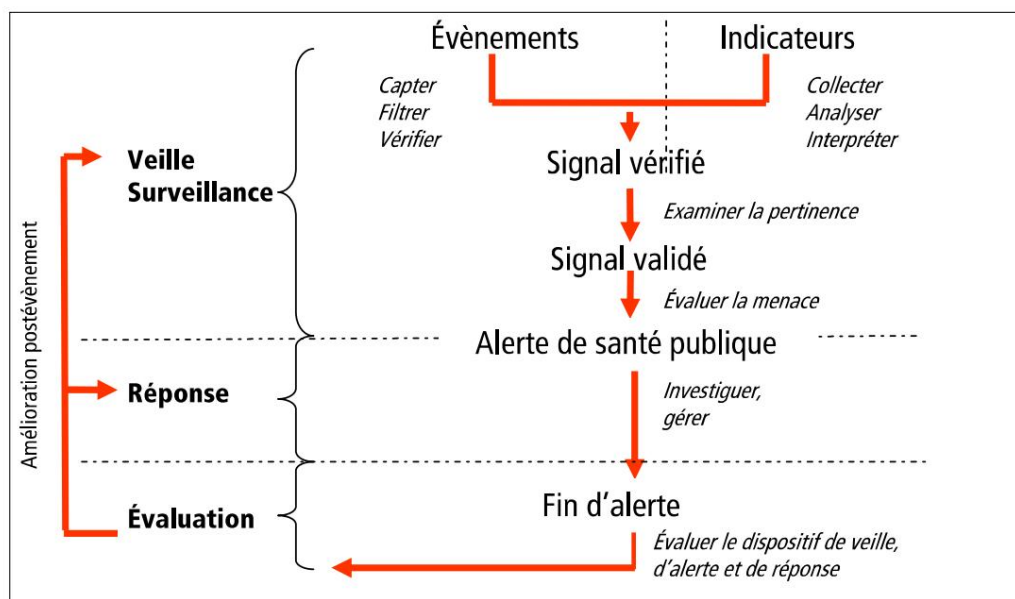
This conceptual framework includes the following actions:

¹² Health monitoring and alert in France, Health Monitoring Institute, 2011 , DGS,
03.11.2014-Updated 06.22.2020 VD HEAT WIND PLAN

- collection of signals of different nature;
- verification of information;
- analysis of health and/or environmental data;
- implementation of etiological diagnostic means;
- investigation of situations, particularly epidemic episodes;
- transmission of alert messages at the appropriate level;
- implementation of health management/response measures;
- evaluation, particularly in the form of feedback.

Schematically, the surveillance system can be illustrated by the conceptual framework below.

below which includes the watch, alert and response phases to a threat.



5.6 INDICATORS

From June 1 to August 31, the DGS receives daily data relating to previously defined indicators. Among this information is the MétéoSwiss bulletin relating to the forecast of maximum temperatures for the next 7 days, as well as a more detailed forecast when a wave

heat is expected (night temperatures, dew point, etc.). This bulletin is subject, if necessary, a more in-depth discussion with the meteorological office.

Health indicators are also collected every day and reported in the ad hoc database, developed by Unisanté. This information concerns:

- visits to pediatric (HEL) and adult (CHUV) emergency rooms, stratified by age groups and by the presence of fever (temperature above 38°C, for admissions to CHUV);
- the number of calls to the CTMG;
- the number of deaths.

All information relating to the indicators is collected and inserted into the database (cf.

image below). This allows you to have an overview of the situation in relation to a standard calculated on the

based on information collected in previous years during the same period and to identify events that do not correspond to situations that can be expected. This database represents a decision support tool on which the DGS relies to assess the need to issue a heatwave warning and, where appropriate, to disseminate information to partners to trigger their heatwave plan.

Date	Semaine	Température (Genève)			Mortalité		Visites Urgences CHUV					Visites Urgences HEL		Appels CTMG
		Min	Moyenne	Max	Totale	≥ 75 ans	Total	≤ 5 ans	5-70 ans	> 70 ans	T>38°C	Total	≤ 5 ans	
03.07.2015	27/2015			35.8	12	8	155	1	115	26	5	97	41	517
04.07.2015	27/2015			37.1	20	10	167	1	121	35	9	125	62	753
05.07.2015	28/2015			38.9	19	13	169	1	124	34	4	117	64	844
06.07.2015	28/2015			33.6	20	11	203	2	145	44	6	82	37	515
07.07.2015	28/2015			39.8	13	7	189	0	139	37	5	85	53	451

Concerning mortality, until the summer of 2015, the DGS asked funeral companies to transmit daily information relating to deaths recorded the day before. Following the observation that this information was difficult to use, because it was not correlated with civil status data transmitted once a week, the DGS abandoned this system.

As of summer 2016, data on mortality have therefore come from a single source, namely the center administrative of civil status with, however, a lag: the deaths of day X are entered in the database of data two weeks after their occurrence. The frequency of transmission of information cannot currently be improved because it represents the shortest time frame allowing the most precise information possible to be obtained. This system is similar to the surveillance system implemented placed by the Federal Statistical Office.

The DGS, however, reserves the possibility, if necessary, to call on funeral companies. to have more information on deaths, under art. 75 para. 5 of the Regulation respecting deaths, burials and funeral directors, September 12, 2012: "In all cases where the interests of health public requires it, the manager of the (funeral) company is required to provide the doctor with cantonal, within the set deadlines, any information or quantified information that it considers useful.

5.7 ALERT LEVELS

The table on the following page illustrates the different alert levels with health responses corresponding. The colors correspond to the thresholds defined in "The general order – Heat wave" adopted by the cantonal conduct control authority¹³.

The transition to a higher stage, or the return to a lower stage, depends on several parameters, also both meteorological and health. Consequently, each stage is declared by the cell responsible for management (DGS and/or EMCC) in collaboration with the partners of the Heatwave Alert Group.

MeteoSwiss also forecasts a danger level of degree 4 corresponding to a HI>93, but it is not not explicitly mentioned as being a specific alert threshold, because it is an integral part of the definition of the warning level (yellow) and the following ones.

¹³ This document is updated annually and can be requested from the EMCC, via its representative within the Group heatwave alert.

ALERT AND HEALTH RESPONSE LEVELS

LEVEL	DEFINITION	HEALTH SITUATION	MEASURES AND MEANS	CONDUCT
DAY BEFORE	HI < 90 (T° < 33°C) Current situation, from June 1 to August 31	Normal Partner standby	• Data collection once a day • Dissemination of advice to partners and the population • Joint preparation	MC Heatwave alert group
CAUTION	HI > 90 (T° 33°-34°C) Forecast and/or measurement for at least 3 consecutive days	Normal or slight increase in health activities upstream of the hospital sector	• Regular contacts with MétéoSuisse and GE • Partner mobilization • Press release(s)	MC Heatwave alert group
HEAT WAVE PERSISTENT	HI > 90 (T° 33°-34°C) Intense heatwave and/or lasting more than 3 days	• Normal or increase in health activities upstream of the hospital sector. • First deaths due to heat	• Triggering socio-health institutional plans (Hospitals, EMS, CMS, ESE, etc.) • Community visits • Strengthening the CTMG • Press release(s), radio/TV - If indicators on the rise: • Infra-ORCA measurements	DSAS MC Heatwave alert group with EMCC
SEVERE HEAT WAVE STATE OF CRISIS	HI > 90 (T° 33°-34°C) for 7-10 days and installation forecast, +/- aggravating factors (pollution, nighttime temperature > 21°, saturation of services, etc.)	• Critical health intervention threshold reached or exceeded (ambulatory services) • Significant increase in hospital activities, possibly saturation • Significant number of deaths	• Health mobilization as well as recall and reinforcement of teams • Community visits - If indicators on the rise: • EMCC switch if the Heat Wave plan is triggered • Reinforcement Visits + Hotline with PCi • Daily press briefings	CE delegation DSAS EMCC (coordination) Heatwave alert group

5.8 HEALTH RESPONSES

The table on the previous page illustrates the alert levels but also the health responses corresponding.

The various socio-health actors who participate in the heatwave system have, as mentioned briefly above, a specific responsibility towards their patients. They must therefore plan specific measures in the event of a heatwave and deploy them in the event of real danger and if the MC has urged them.

The chapters below explain certain provisions as a health response in the event of severe heatwave.

5.8.1 STRENGTHENING THE CTMG

When the situation reaches the orange alert level (persistent heatwave), the mission of the CTMG consists of:

- provide advice in line with prevention measures, centralize questions, and standardize responses,
- quickly inform and direct people with symptoms or risk criteria, or requiring the “visitors” service (see below), to the most appropriate health services or to the municipalities (so that they can take into account the wishes of people to benefit from prevention services – cf. 5.8.2), this as far upstream as possible of emergencies,
- advise community visitors based on the situation they discover at home,
- raise the concerns of the population and professionals to the management unit crisis.

The CTMG telephone number is clearly highlighted in the prevention material. In case to reach the orange level, the FUS, which manages the CTMG, calls on a number of personnel extra trained – even if briefly. If necessary, questions and answers which give the main instructions adapted to different audiences, can be developed in addition to the DGS leaflets.

Due to the presence of many tourists in summer, in addition to non-French-speaking residents, it is necessary to look for backup telephone operators capable of speaking the foreign languages most represented in the region.

The role of the CTMG concerns the orientation of people, the assessment of symptoms, the mobilization of medical care, and supervision of visitors in real time, as well as the capacity to alert about a saturation of services upstream of accommodation. In addition to activating the hotline, the strengthening of the CTMG/144 provides for an increase in the number of advice-guidance lines medicalized and the mobilization of picket teams.

The CTMG also receives “Sécutél” type calls (automatic alarm for the elderly). or dependents living at home), likely to concern one of the main target populations in case of severe heatwave.

5.8.2 COMMUNITY VISITORS

The protection of vulnerable people is the primary responsibility of municipalities. The municipalities were chosen for reasons of proximity and knowledge of their population, data protection and acceptability of visits. The DGS intervenes to support this role of common, particularly through the definition and dissemination of prevention messages to the general public.

The aim of this approach is to constitute a first line of prevention of the consequences of heat on health. It is also a matter of detecting the first symptoms before the target audience requires specialized care - knowing that complications can be quickly fatal, most often before arriving at the hospital. This approach should make it possible to limit the number of avoidable deaths at home, but also to protect downstream health services (overload of services with possible consequences on intra-hospital morbidity and mortality) by allowing them to function normally without harming other patients.

The process includes, for municipalities, a preparatory stage of identifying people vulnerable people not known to CMS, and recruitment/training of community visitors. A second step consists of mobilizing and coordinating these visitors for concrete activities with the people at risk (visits and/or telephone calls).

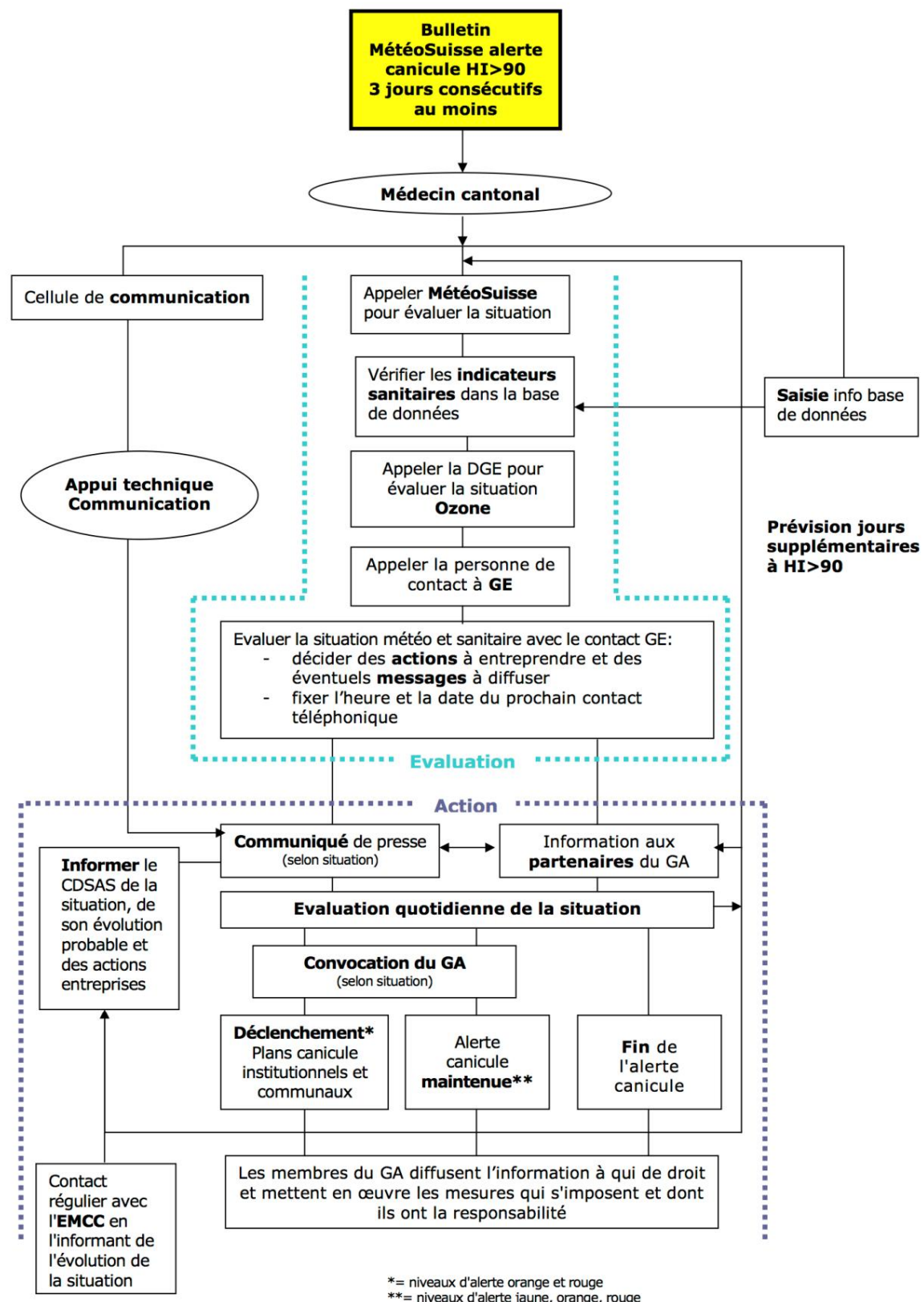
Concerning the implementation of this system, the municipalities must establish, in the spring, the list of their visitors and these, in case of need during the summer, must be able to be engaged with a brief deadline. Preference should be given to municipal collaborators and people known in the municipality through their voluntary investment in other activities.

The establishment of PCi duty-bearers should not be considered to ensure the management of the first days of the heatwave (set-up time and training). The municipalities, which will have to resort to support of the PCi, will ensure that ORPC commanders are involved early enough to enable them to develop their personnel planning.

Appendices 1 to 3 of this document detail the procedures for the attention of municipalities for the implementation of visitor arrangements.

6. THE PROCESS

The diagram below illustrates the process to follow when MeteoSwiss issues a heatwave alert.



7. COMMUNICATION STRATEGY AND MATERIAL

The communication strategy is essential in any public health action so that it has every chance of success.

As a reminder, the main objective of the Heatwave Plan is to reduce morbidity and mortality rates related to heat. To do this, the mission of the communication campaign is to:

- disseminate prevention messages aimed at protecting vulnerable populations against the health consequences of the heatwave (elderly people, chronically ill people, children under age, workers from certain professional backgrounds);
- inform professionals in the socio-health sector about:
 - preventive measures to be provided in establishments;
 - curative measures to be adopted in the event of a heatwave;
- inform the population, partners and socio-health professionals of the measures taken by the State at each phase: watch, alert and crisis;
- inform partners about DSAS's expectations regarding their role within the framework of the Plan heat wave.

Each year, the DGS defines the detailed communication schedule as well as the support (leaflets and posters) to be used for the dissemination of information to different target audiences. This material can be consulted and ordered at www.vd.ch/canicule.

New means of disseminating information are investigated and adopted if deemed appropriate: presence on Facebook, TV screens at post offices, etc.

8. EVALUATION OF THE PROCESS

At the end of each heat wave health monitoring period, a session is organized with the participation of all partners in order to establish an assessment, exchange experiences, discuss the positive and/or negative aspects in order to make possible corrective measures during the monitoring period next.

The main points to be addressed and evaluated mainly concern:

- the weather situation (was there a heatwave? For how many days, what temperatures?);
- health indicators;
- management of surveillance by the DGS;
- partner experiences;
- communications strategy and materials.

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APPENDICES

- Annex 1:** **Organization of visitors within the framework of the Heatwave Plan: Implementation procedure for the attention of municipalities**
- Appendix 2:** **Standard specifications for visitors**
- Appendix 3:** **Form for visitors**

ANNEX 1 IMPLEMENTATION PROCEDURE FOR THE ATTENTION OF MUNICIPALITIES

The objective of this document is to provide a common thread to municipalities for the development and implementation of a heatwave system at the municipal level. It is not binding.

1. Cantonal organization

has. Objectives of the heatwave plan at cantonal level

The cantonal objectives of the system are to: ÿ

Prevent or mitigate health damage due to extreme heat, while ensuring continuity with primary prevention activities (information for professionals and the population, new brochures and posters developed by the DGS, early detection of symptoms).

ÿ Avoid overloading the health system (home care, medico-social accommodation and hospital sector) by controlling patient flows upstream.

ÿ Prepare, mobilize and coordinate all stakeholders involved in the response health, including municipalities.

b. Management structure at cantonal level

ÿ Overall responsibility for the heat wave system lies with the head of the Department of health and social action (DSAS).

ÿ The overall management of the heatwave system is the responsibility of the Doctor cantonal.

ÿ In the event of a deterioration of the situation due to associated event(s) such as drought, energy crisis, major fires, the head of the cantonal management staff (EMCC) can trigger the general plan of "Heat wave" coordination and appropriate measures.

vs. Heatwave watch and alarm device

A heatwave monitoring system is provided annually by the Directorate General of Health (DGS) from June 1 to August 31. If the alert threshold is reached, the Prefect, informed by the DGS, will notify the municipalities of the situation and ask them to activate their heatwave plan as quickly as possible. Based on the goodwill of the municipalities, only those who have sufficient preparedness will be able to contribute effectively to this operation.

2. Municipal organization

has. Objectives of the plan at the municipal level The

municipal objectives of the system are to: ÿ Prevent or

limit the harmful effects of heat on the health of vulnerable people identified within the municipality, not known to the Medical-Social Centers (CMS), ensuring regular contact with people at risk.

ÿ Disseminate heat protection instructions.

ÿ Identify people suffering from the effects of the heatwave and/or requiring medical care. ÿ Transmit essential information to the

Canton, via the Prefect-Prefect, concerning the implementation of the plan and its effects.

b. Basic principle This

involves:

ÿ Identifying potentially vulnerable people within the municipality. ÿ To offer them the possibility of follow-up in the event of a heatwave, by contact telephone and/or by home visit with dedicated staff.

ÿ To ensure reciprocal information between municipalities and Canton.

Even if the basic principle of implementation is similar for all municipalities, it nevertheless involves taking into account the density of the population at risk in the municipality and adapting the dedicated resources and its organization accordingly. Large cities (urban areas) are obviously the municipalities where the risk is the highest and require a high state of preparation for this purpose, in terms of organization and implementation of dedicated resources.

vs. Municipal management

In a manner similar to the municipal organization for economic supply, a team to manage the system and its preparations must be set up, made up of at least a representative of the municipality and an administrative and operational respondent. .

3. Planning action (preparatory measures at municipal level)

has. Period

March 1 to May 31

b. Planning steps

1. Prospective census of the population at risk
2. Information for the population concerned and offer of follow-up services 3. Recruitment of staff dedicated to "visit" follow-up services
4. Staff training and establishment of a commitment schedule 5. Communication

Step 1 Census of the vulnerable population

This step consists of establishing a municipal reference list of people over 75 years of age not known to the CMS.

Procedure

- ÿ Extract from the inhabitants control the list of people who on June 1st of the current year will be at least 75 years old as well as their contact details.
- ÿ Contact the CMS attached to the municipality and send them the list established for cross-checking with its data.
- ÿ The CMS returns the "corrected" list to the municipality, in which the people known to the health services were excluded.
- ÿ The list of the potentially vulnerable population is established and serves as a reference base for the rest of the process. It should be kept in all cases.

Step 2 Information for the population concerned and offer of follow-up services

This step consists of sending an information letter (or the publication of information in the entire municipal household) accompanied by a response bulletin for the attention of potentially vulnerable people and aims to establish a list community of vulnerable people requesting a follow-up service during heat waves.

From this stage, the procedure must be brought to its conclusion, so as to guarantee the services resulting from it, that is to say the implementation of a visit system

Procedure

- ÿ Sending mail with municipal header and footer, signature and response time. ÿ Sending according to defined list with reply envelope already stamped. ÿ Sorting return mail and refining the list of the vulnerable population by excluding people who do not wish to be monitored by the municipality. ÿ Establish a list of vulnerable people requesting a monitoring service on the basis census results.

Step 3 Staff dedicated to "visit" monitoring services

This step consists of defining the resources necessary for "visit" monitoring services based on the number of vulnerable people announced and provides an overview of possible recruitment methods.

Procedure

- ÿ Based on the results of the census, define the number of staff to be recruited in function: -
 - the number of people requesting home monitoring, - the frequency of visits (daily during prolonged heatwave periods ÿ3 days), - the territorial configuration of the municipality (private or public mobility necessary).
- ÿ Define with what type of personnel the municipality is able to fulfill its service (existing municipal resource, including PCi / volunteers / auxiliaries, hired for a 28

- fixed duration in case of implementation of the plan)
- ÿ Carry out a criminal record check of staff engaged as volunteers or auxiliary.

Step 4 Training of dedicated staff

This step consists of preparing the staff dedicated to their tasks and setting the schedule for everyone's availability and commitments in relation to the "visit" monitoring services to be provided in the event of a heatwave (see standard specifications at the page 4).

Procedure

- ÿ Bring together visiting staff for an information session. ÿ Provide training in collaboration with a CMS representative attached to the municipality or the Samaritans.
- ÿ Establish a schedule of staff availability and carry out a distribution of vulnerable people to follow.

Step 5 Organization of visits

This stage consists of ensuring the bond of trust between the staff and the people who request to be monitored, and of organizing contacts and visits.

Procedure

- ÿ Inform beneficiaries and send them information on dedicated and authorized staff to provide support (name and ideally photo portrait). ÿ Establish contact with beneficiaries and dedicated staff through telephone contact or a prior visit.
- ÿ Facilitate the identification of dedicated personnel by providing a uniform or official paper which justifies its activity.

ANNEX 2 STANDARD SPECIFICATIONS FOR VISITORS

Missions	• Prevent or limit the harmful effects of heat on health among vulnerable people identified by the municipalities, not known to the CMS, by guaranteeing regular contact with people at risk • Disseminate heat protection instructions • Identify people suffering from the effects of the heatwave and/or requiring medication in medical care
Tasks	• Contact by telephone the people identified by the municipalities or who are spontaneously announced • Keep the lists provided by the municipalities up to date • Establish simple daily statistics of the services provided • Carry out the visit on the basis of a checklist provided by the DGS ("visitors" flyer)
Supervisor	Administrative tasks: Trustee (and municipal services) Medical questions: attending physician or CTMG Overall responsibility: SSCM
Workplace	Municipal house (rear base) and clients' homes • Mobile phone •
Material	Non-sterile disposable gloves • Client lists and pens • Thermometer (apartment temperature) • Identification badge • Leaflets/Flyers provided by the DGS: "visitors", "Heatwave: stay cool" and "A health problem? » • Vehicle • Parking permit • Light clothing and hat
Specific Skills	• Interpersonal skills, sociability, good personal presentation • Practical sense and resourcefulness • Sense of service to others and interpersonal skills • Driving license • Basic health knowledge an asset • Basic computer knowledge • Knowledge of the municipality (location of addresses, resources and services available, stores and places refreshed, etc.) • Availability at very short notice for at least one of the 3 months from June to september
Desired profiles	• Samaritans • Red Cross auxiliaries and volunteers • PCI Staff • Other volunteers > 18 years old (parishes, various societies)

ANNEX 3 FORM FOR USE BY VISITORS



Direction générale de la santé

Bâtiment administratif de la Pontaise
Avenue des Casernes 2
CH - 1014 Lausanne

T +41 21 316 42 00
F +41 21 316 42 78
info.santepublique@vd.ch
www.vd.ch/dgs

CANICULE

FORMULAIRE VISITES À DOMICILE

DIRECTIVES AUX PERSONNES SE RENDANT AU DOMICILE

En prévision de la période de surveillance canicule, les communes vaudoises sont invitées à mettre en place un système de visiteurs communautaires dont le but est de constituer une première ligne de prévention des conséquences de la chaleur sur la santé. Il s'agit également de détecter les premiers symptômes avant qu'ils ne nécessitent une prise en charge médicale.

L'ABC DU VISITEUR

- A. Avant la visite: appeler la personne pour l'informer de la visite
- B. En prévision de la visite: se prémunir d'un thermomètre pour la mesure de la chaleur dans le logement
- C. À l'arrivée: préciser son identité, montrer son badge et expliquer l'objectif de la visite

PERSONNE VISITÉE

Sexe: ☐ Homme ☐ Femme

Nom: Prénom:

Adresse (Rue/NPA/Localité):

Date de naissance: / /

Tél. fixe: Tél. mobile:

Personne de confiance et n° tél.:

POINTS À VÉRIFIER

- | | | |
|---|------------------------------|------------------------------|
| 1. Êtes-vous en contact régulier avec vos proches ou vos voisins (1x/jour)? | ☐ Oui | ☐ Non |
| 2. Pensez-vous à protéger votre logement de la chaleur (rideaux, volets fermés, stores baissés, parasols, etc.)? | ☐ Oui | ☐ Non |
| 3. Quelle température fait-il dans le logement? | <input type="text"/> | degrés °C |
| 4. Votre réfrigérateur est-il en bon état de marche? | ☐ Oui | ☐ Non |
| 5. Buvez-vous régulièrement? | ☐ Oui | ☐ Non |
| 6. Portez-vous des habits légers? | ☐ Oui | ☐ Non |
| 7. Connaissiez-vous les mesures de prévention en cas de canicule? | ☐ Oui | ☐ Non |
| 8. Avez-vous mis les coordonnées téléphoniques de votre médecin traitant et de la Centrale téléphonique des médecins de garde 0848 133 133 près du téléphone? | ☐ Oui | ☐ Non |



CANICULE

FORMULAIRE VISITES À DOMICILE

SI NON À UNE OU PLUSIEURS QUESTIONS

Faire le nécessaire avec la personne pour tenter de corriger la situation. Inscrire les actions entreprises ou à entreprendre dans le journal de visite à la page 3 de ce formulaire.

INDÉPENDAMMENT DES RÉPONSES FOURNIES À CES QUESTIONS

- Répéter les consignes de protection de la chaleur et de la santé: passer en revue et remettre les flyers «Canicule: rester au frais» et «Un problème de santé?»¹
- Demander à la personne de donner de ses nouvelles par téléphone au moins une fois par jour à son entourage familial si existant (ou au numéro de téléphone éventuellement prévu)

RECHERCHE DES SIGNES D'ALERTE DUS À LA CHALEUR

9. Depuis les fortes chaleurs ou la canicule, avez-vous:

- | | | |
|---|------------------------------|------------------------------|
| a. des troubles du sommeil inhabituels? | ☐ Oui | ☐ Non |
| b. des maux de tête violents, des vertiges inhabituels? | ☐ Oui | ☐ Non |
| c. une sensation inhabituelle de fatigue, de faiblesse, de malaise? | ☐ Oui | ☐ Non |
| d. des nausées ou des vomissements ou des diarrhées? | ☐ Oui | ☐ Non |
| e. des difficultés inhabituelles à vous déplacer? | ☐ Oui | ☐ Non |
| f. de la fièvre supérieure à 38.5°C? | ☐ Oui | ☐ Non |
| g. des crampes? | ☐ Oui | ☐ Non |

10. La personne semble-t-elle avoir des propos confus, incohérents? ☐ Oui ☐ Non

SI OUI À UNE OU PLUSIEURS QUESTIONS OU EN CAS DE DOUTES

- Appeler le médecin traitant ou la centrale téléphonique des médecins de garde au 0848 133 133
- En cas d'urgence vitale, appeler le 144

EN ATTENDANT LES SECOURS

- Si la personne est consciente et lucide: lui donner à boire de l'eau fraîche
- Rafraîchir la personne avec des linges humides ou avec un brumisateur
- Utiliser un ventilateur si disponible
- **Attention:** si la personne perd connaissance, la mettre en position latérale de sécurité.

¹ Le matériel de présentation peut être consulté à l'adresse www.vd.ch/canicule et commandé gratuitement à l'adresse info.santepublique@vd.ch



CANICULE

FORMULAIRE VISITES À DOMICILE

JOURNAL DES VISITES			
DATE	RÉFÉRENT	ACTION (TÉL./VISITE/AUTRE)	REMARQUE/SUIVI/ANOMALIES CONSTATÉES (PRÉCISER LE N° DE LA QUESTION À LAQUELLE LA REMARQUE SE RÉFÈRE)

